

PULSE

LFO TREMOLO & FILTER



What Pulse does

A shape-selectable LFO that chops your signal and sweeps a filter, free-running or locked to the host.

Pulse runs one low-frequency oscillator and uses it for two jobs at once. It modulates **amplitude** – the classic tremolo or rhythmic gate – and it sweeps the cutoff of a **resonant low-pass filter**. Both are driven by the same wave, so they always move in step with each other and with the beat.

The LFO either free-runs in hertz or locks to the host tempo and transport. When locked, its phase is pinned to the playhead, so a 1/8-note gate lands on the eighth notes no matter where you start playback.

Installing

Run the installer, [Pulse-1.2.1.pkg](#). At the destination step, choose whether to install for **everyone who uses this Mac** or **just for you**. Installing for everyone asks for an administrator password; installing for yourself does not, and keeps everything inside your home folder.

FORMAT	EVERYONE	JUST ME
Audio Unit	/Library/Audio/Plug-Ins/Components	~/Library/Audio/Plug-Ins/Components
VST3	/Library/Audio/Plug-Ins/VST3	~/Library/Audio/Plug-Ins/VST3
Standalone	/Applications	~/Applications

You can also choose which of the three to install. All are universal binaries, running natively on both Apple Silicon and Intel Macs. Restart your DAW afterwards so it rescans for new plug-ins.

Reading the interface

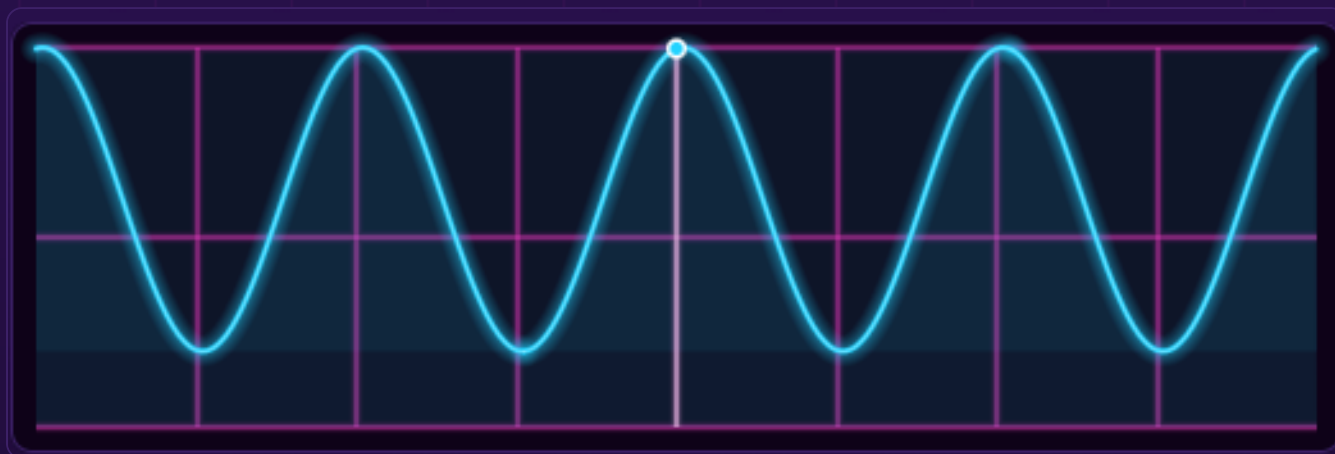
Colour tells you which section a control belongs to:

- **Magenta** – the shape of the modulation: wave, width, rate.
- **Electric blue** – amplitude, and its curve in the display.
- **Yellow** – the filter, and its curve in the display.

Every knob responds to click-and-drag, and to scrolling. Double-click a knob to return it to its default. The value readout under each knob is editable – click it and type an exact figure.

Watching the modulation

Time runs left to right. The centre line is now.



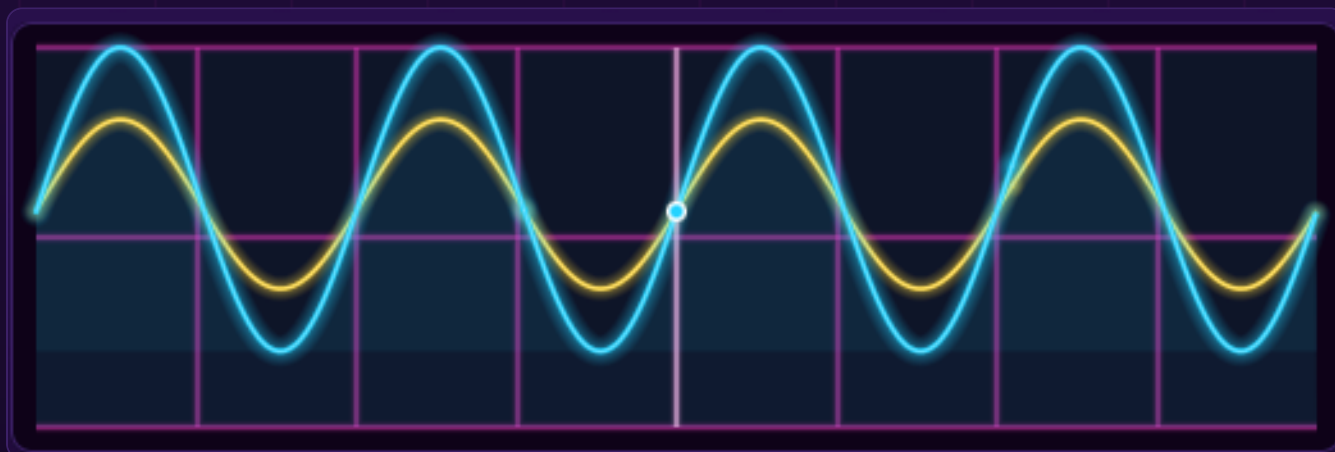
The amplitude envelope, drawn in electric blue. Left of the centre line is what has already played; right of it is what is about to.

The display does not show your audio – it shows the **modulation being applied to it**. The vertical axis is how open the signal is: the top of the plot is fully open, the bottom is fully closed. The white dot on the centre line marks the value being applied at this instant.

Because the curve is generated from exactly the same maths the audio engine uses, what you see is what you hear – including the way the wave will look several cycles from now.

The second curve

As soon as the filter is doing something, a **yellow curve** joins the blue one. It plots the filter's cutoff on the same vertical axis: high on the plot means the filter is open and bright, low means it is closed and dark.



Both curves at once. The yellow filter line rises and falls with the blue amplitude line – you can see the filter opening on every swell.

TIP

If the yellow line is missing entirely, the filter is fully bypassed: the envelope is at zero *and* cutoff is wide open. In that state Pulse passes the signal through untouched.

Choosing a shape

Six shapes, from smooth swells to hard gating to random steps.



Click a tile to select it. The lit tile is the active shape.

SHAPE	CHARACTER
Square	Hard on/off gating. The only shape that uses Width. Edges are smoothed by about a millisecond so they click rather than crack.
S&H	Sample & hold – a new random level each cycle, held flat. Stepped, unpredictable movement. The sequence is fixed, so it plays back identically every time.
Sine	Smooth, symmetrical swelling. The classic amp tremolo.
Triangle	Linear up and down. Similar to sine but with a firmer turn at the peaks.
Saw Up	Ramps open, then snaps shut. Reverse-swell, backwards-tape feel.
Saw Down	Snaps open, then decays away. Plucked and percussive.

Width

Width sets the duty cycle of the square wave – the proportion of each cycle spent open. At 50% you get an even on/off chop. Wind it down for short stabs, up for brief gaps in an otherwise open signal.

It has no effect on the other five shapes, and greys out to tell you so.

NOTE

Sample & hold changes level once per cycle rather than continuously, so with it selected Rate controls how often a new random value is chosen.

Rate and depth

How fast the wave runs, and how hard it bites.



Rate

In free mode, Rate sweeps from **0.05 Hz** — one cycle every twenty seconds — up to **20 Hz**, well into audible-flutter territory. The scale is exponential, so the slow end gets as much of the knob's travel as the fast end.

Sync

Engage **SYNC** and Rate switches from hertz to note divisions, running from **1/1** down to **1/32** and taking in dotted and triplet values on the way. The readout under the knob shows the current division.

Synced, the LFO locks its phase to the host playhead, so cycles land on the grid wherever you drop the cursor. If the transport is stopped, it free-runs at the equivalent tempo rather than freezing.



Amplitude

Amplitude is the depth of the tremolo — how much of the wave's travel is applied to the signal.

- At **0%** the signal passes untouched, whatever the wave is doing.
- At **50%** the level swings between full and half.
- At **100%** the troughs reach silence. With a square wave, that is a hard gate.

The shaded band behind the curve in the display shows the range the level is sweeping through, so you can see the depth as well as hear it.

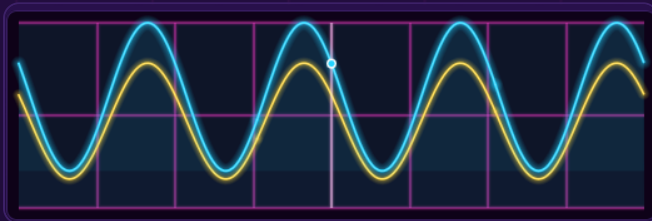
Sweeping the cutoff

A resonant 12 dB/octave low-pass, driven by the same LFO.

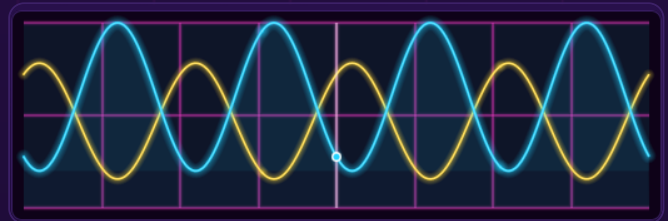


Filter — the envelope amount

This control is **bipolar**. It sits at zero straight up, and its ring fills outwards from twelve o'clock rather than from the left, so "off" reads as an empty ring at a glance.



Positive. The filter opens with the wave – bright at the peaks, dark in the troughs. The yellow curve tracks the blue one.



Negative. Inverted – the peak of the wave closes the filter. The yellow curve mirrors the blue one.

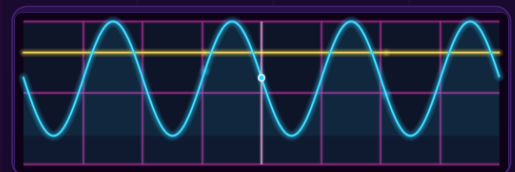
Cutoff and Resonance

Cutoff is the ceiling the envelope sweeps down from, adjustable between **80 Hz** and **20 kHz**. At full travel the filter is wide open; lower it and the whole sweep moves down with it. With the envelope at zero it simply acts as a static low-pass, which is a useful way to tame a source before the tremolo hits it.

Resonance lifts a peak at the corner frequency, from flat at 0% to a sharp, vocal squelch at 100%. Combined with a slow envelope it gives you an auto-wah; with a fast one, a synth-like bubble.

Cutoff on its own

With **Filter** at zero, nothing sweeps the cutoff and the yellow line in the display flattens out. What you have left is a plain static low-pass – a useful way to tame a bright source before the tremolo reaches it.



WATCH YOUR LEVELS

High resonance adds considerable gain around the cutoff – up to about 20 dB at the extreme. On bass-heavy material, pull the source down before pushing Resonance up.

Every parameter

Ranges and defaults. All eight are automatable from the host.

CONTROL	RANGE	DEFAULT
Wave	Triangle · Saw Up · Saw Down · Square · Sine · S&H	Sine
Rate (free)	0.05 – 20 Hz, exponential	1.00 Hz
Rate (synced)	1/1 · 1/2 · 1/2T · 1/4 · 1/4T · 1/8 · 1/8T · 1/16 · 1/16T · 1/32	1/4
Host Sync	Off / On	Off
Width	1 – 99 % (square wave only)	50 %
Amplitude	0 – 100 %	80 %
Filter	–100 – +100 % (0 = off)	0 %
Cutoff	80 Hz – 20 kHz	20 kHz
Resonance	0 – 100 % (Q 0.7 – 10)	0 %

Starting points

Classic amp tremolo

Sine · free · 4–6 Hz · Amplitude 40% · Filter 0%. Gentle, continuous swelling. Take Amplitude down to 20% for something that sits under a vocal without drawing attention.

Rhythmic gate

Square · sync 1/8 or 1/16 · Amplitude 100% · Width 50%. A hard chop locked to the grid. Narrow Width to around 25% for shorter, more percussive stabs.

Auto-wah

Sine · sync 1/2 · Amplitude 0% · Filter +70% · Cutoff 6 kHz · Resonance 60%. Amplitude at zero leaves the level alone, so the filter sweep is all you hear.

Pumping

Saw Up · sync 1/4 · Amplitude 70% · Filter +40% · Cutoff 10 kHz. Ducks on the beat and recovers across it, brightening as it returns – the sidechain-compressor trick without a sidechain.

Broken machine

S&H · sync 1/16 · Amplitude 60% · Filter –80% · Resonance 75%. Random levels with the filter slamming shut on the loud steps. Chaotic, and different in feel every bar.